



Fire-Guard Robot

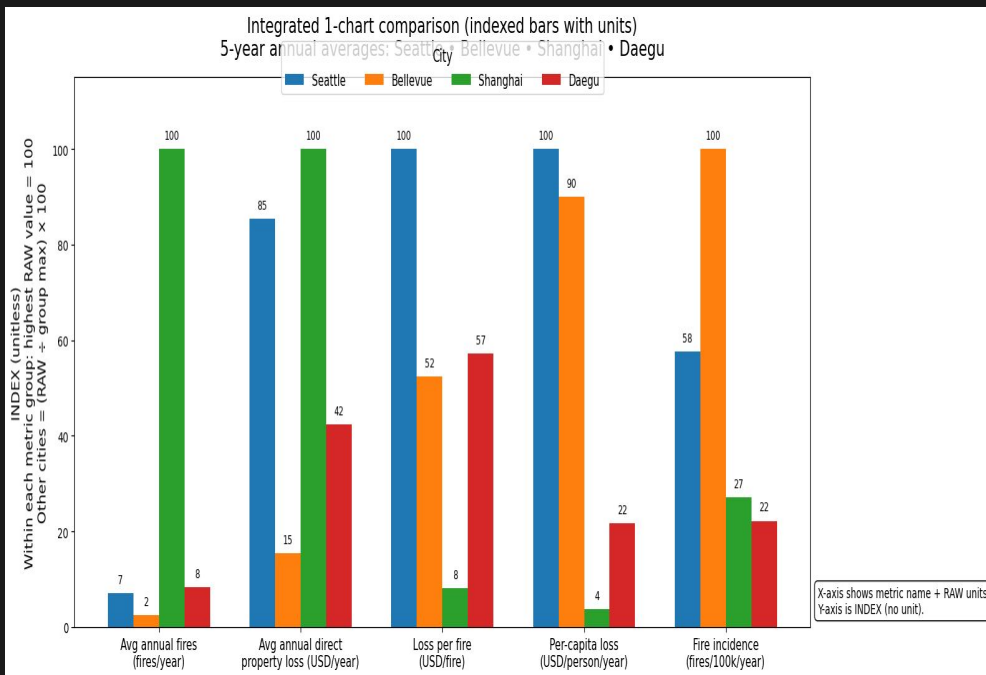
Saving Firefighters' Lives

"Better to melt a robot than burn a firefighter."

TEAM 9

Fire Risk is Local & Global

Comparative Fire Statistics: Bellevue vs. Global Cities



1. Highest Fire Risk

2. High Financial Impact

3. Severe Event Intensity

4. The Solution: Speed

• Conclusion

- We face **Frequent Fires & High Costs**.
- **Early detection** is not an option, it is a necessity for our city.

**We cannot change the statistics overnight,
But Fire-Guard can change the outcome.**

Target Persona



Profile

- Name: Captain James
- Role: Incident Commander
- Experience: 15+ Years
- Goal: "Bring everyone home safe."

His Pain Point

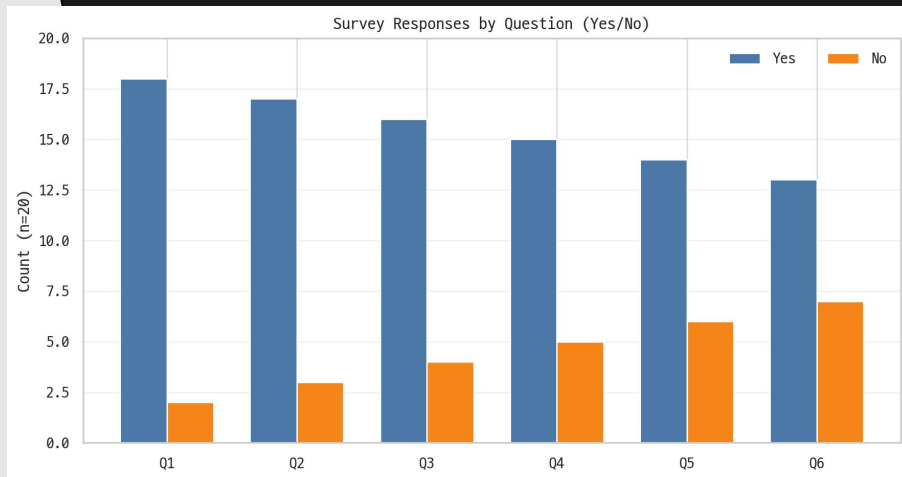
- "Sending my team into a black box is my biggest fear."
- He has to make life-or-death decisions within seconds, but currently, he relies on intuition, not data.

Citizen interviews

20 people were surveyed.

● Question

1. Do you agree that firefighters operate in highly hazardous conditions? (Yes / No)
2. Did you know that firefighters often crawl into smoke-filled buildings blindly, unable to see their own hands? (Yes / No)
3. Do you agree that a robot should enter a burning building *before* a human firefighter to check for victims? (Yes / No)
4. Do you think seeing a real-time map of the fire would help firefighters make better decisions than just guessing? (Yes / No)
5. Based on this, do you support the Bellevue Fire Department adopting these low-cost "Expendable Scout" robots? (Yes / No)
6. If this equipment were funded by donations instead of taxes, would you be willing to contribute? (Yes / No)



Fire-Guard



Fire-Guard

- Today



- Tomorrow

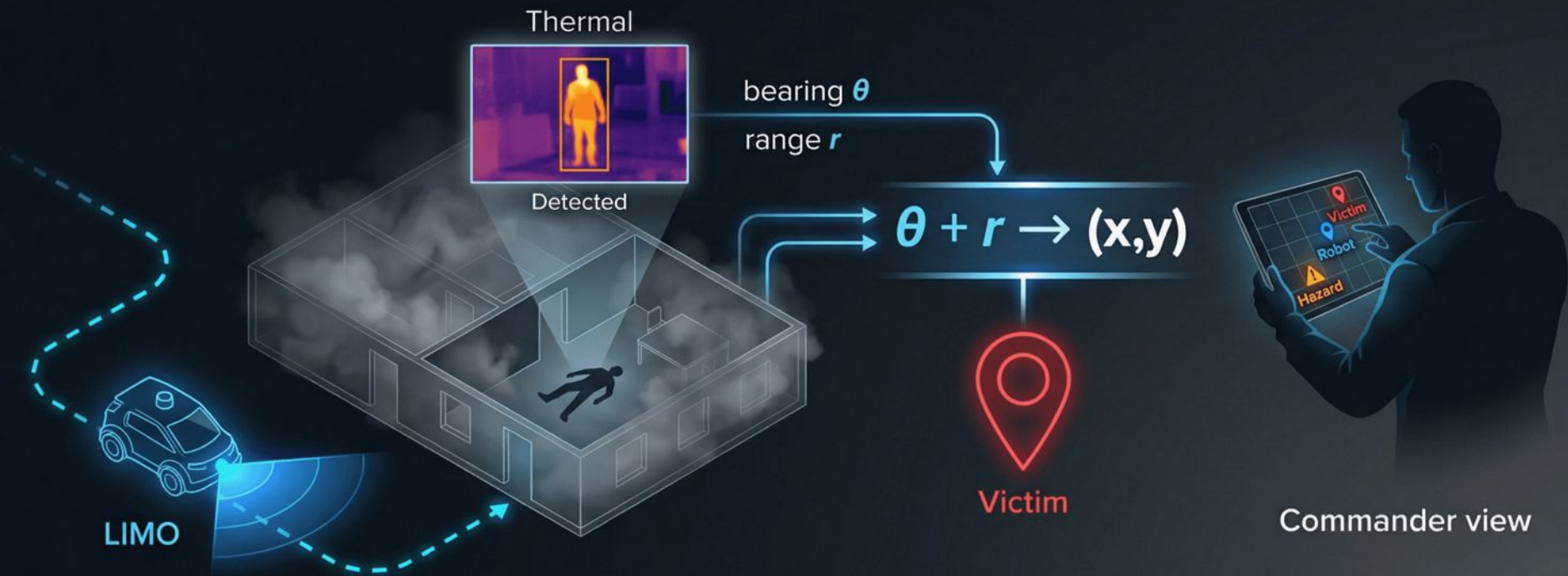
Material upgrade

- Flame-Retardant Skin
- Heat-Resistant Shell
- Industrial Durability

High-rise building solutions

- Drone Integration
- Air-Ground Team

Field-to-Map Workflow



Incident Commander Dashboard



Fire-Guard vs Teledyne FLIR PackBot 510

Feature	Teledyne FLIR PackBot 510 (Competitor)	FireGuard (Our Solution)
Primary Role	EOD & HazMat	Rapid Fire Scouting
Unit Price	\$150,000 ~ \$200,000+	Under \$3,000
Weight	14.5 kg (32 lbs) (Too heavy to carry)	3 ~ 5 kg (6-10 lbs) (Hand-portable)
Mobility	Heavy Tracks (Slow deployment)	Agile 4WD/Tracks (Throw & Go)
Key Equipment	Robotic Manipulator Arm (Heavy)	Thermal & LiDAR Only (Lightweight)
Loss Risk	Critical Asset (Must be recovered)	Expendable Tool (Designed to melt)

Voice of Customer

Visibility over Durability

A device that cuts through smoke and identifies human heat signatures instantly.

Expendability is Key

Low-cost assets that remove the psychological burden of equipment loss.

Speed of Deployment

Ready to deploy in under 60 seconds.
Throw and go.

Tech Specs & Cost Structure

Component Group	Specific Model / Specification	Cost (Est.)
Mobile Platform	AgileX LIMO Base (Ackermann Steering, High Torque)	\$1,800
Thermal Vision	FLIR Lepton 3.5 (Radiometric Temp. Sensing)	\$350
Mapping Sensor	RPLIDAR A2M12 (360° SLAM, 12m Range)	\$150
Main Computer	Raspberry Pi 5 (8GB RAM, ROS 2 Native)	\$100
Material (Shell)	Flame-Retardant Polycarbonate (FR-PC)	\$100
Power & Misc	High-Capacity Li-Po Battery, Cables, PCBs	\$230
TOTAL	System Total	\$2,730

Pilot Roadmap & Success Metrics

Swarm Strategy

-  5-unit starter kit vs 1 expensive robot
-  5x coverage area in same timeframe
-  Redundancy: System survives if one unit fails
-  Risk distribution across multiple assets

3-Month Pilot

-  1 Department deployment
-  5 Units in field testing
-  Real incident validation
-  SOP development & training

Pilot Success KPIs

< 2 min

Time-to-Map

80%+

Victim Detection

-30%

IC Decision Time

Scaling Plan

Phase 2 Months 4–6 · 50–75 Units → Phase 3 Year 2 · 500+ Units

Ready to Pilot?

Request 5-unit starter kit & procurement-ready proposal

Problem & Paradox

The Information Gap

-  Zero visibility in smoke-filled structures
-  Victim location remains uncertain
-  Structural collapse risks unknown
-  Time pressure forces blind decisions

The Deployment Paradox

-  Current robots: \$40k–\$150k+ per unit
-  Too valuable to risk in hazardous zones
-  Equipment sits unused when needed most

Solution

Fire-Guard Specs

-  **Cost:** Under \$3,000 per unit
-  **Weight:** 3–5 kg (throw-and-go)
-  **Sensors:** Thermal + LiDAR
-  **Safety:** Safe Mode when mapping degrades

Competitive Advantage

Feature	Traditional	Drone	Fire-Guard
Cost	\$40k+	\$8k–\$15k	< \$3k
Indoor Smoke	⚠ Limited	✗ No	✓ Yes
Deployment	✗ Hesitant	⚠ Medium	✓ Eager
Victim Detection	✓ Yes	✗ Aerial only	✓ Ground

“Better to melt a robot than burn a firefighter.”

\$3,000 vs One Life

Not only a technology problem—a deployment and priority problem.

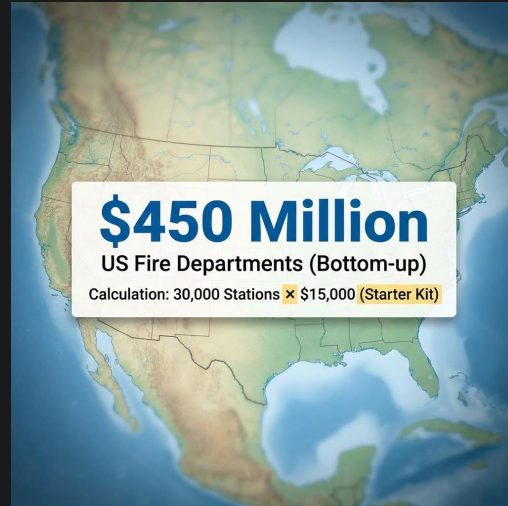
Which do you choose: A melted robot or a
burned firefighter?

Which is more important: A **\$3,000 loss** or a
human life?

Which do you want: **Heroic sacrifice** in
uncertainty or **safe rescue** based on data?

"Let's not repeat the tragedies of the past.

Join us in a new era where robots go first,
and firefighters only enter when it's certain."



Introducing our team

Hyunwoo Jung – Electrical Engineering

Hyeonil An – Electrical Engineering

Wontaek Jung – Mechanical Engineering

Kwanwoo Kim – Mechanical Engineering

Minwoo Nam – adviser

Michael Guo – Electrical Engineering

James Niu – Computer Science Department

Leo Ngo – Environmental Engineering

Frank Lee – adviser



Thank you!!